

ANSWERS

1.

Question	Answer	Marks
2(a)	change / increase of velocity (with time)	B1
	change / increase of velocity per unit time or rate of change / increase of velocity	B1
2(b)(i)	force $\times$ distance moved <u>in the direction of the force / forwards</u>	B1
2(b)(ii)	(work done $\Rightarrow$) $5.4 \times 10^4 \times 280$	C1
	1.5×10^7 (J)	A1

Question	Answer	Marks
2(b)(iii)	$E_k = \frac{1}{2}mv^2$ or ($m \Rightarrow$) $2 \times E_k / v^2$	C1
	($m \Rightarrow$) $2 \times 3.2 \times 10^6 / 12^2$ or $2 \times 3.2 \times 10^6 / 144$	C1
	4.4×10^4 (kg)	A1

2.

Question	Answer	Marks
1(a)	speed is constant for 10 s / at first	B1
	deceleration - has (average) value 0.39 m/s^2 - lasts for 40 s - is non-uniform	B1
1(b)(i)	friction (with road) and <u>air</u> resistance or (air) drag	B1
1(b)(ii)	resistive force(s) change / decrease (as speed decreases) or (graph shows) deceleration is not constant / decreases or non-uniform decrease in speed	B1
1(c)(i)	(momentum $\Rightarrow$) mv in any form algebraic or numerical	C1
	8400 and kg m / s or Ns	A1
1(c)(ii)	($F \Rightarrow$) $\Delta mv / t$ or $F = ma$ in any form algebraic or numerical	C1
	840 (N)	A1

3.

Question	Answer	Marks
7(a)	acceleration / increase in speed and gravity / weight mentioned	B1
	air resistance / friction increases (with speed)	B1
	terminal / constant velocity reached or air resistance (becomes) equal to accelerating force	B1
7(b)	(distance travelled =) area under the graph or $(x =) \frac{1}{2}(u + v)t$ or $\frac{1}{2} \times 15.2 \times 4.0$	C1
	$35 - (\frac{1}{2} \times 15.2 \times 4.0)$ or 30.4 or 30	C1
	4.6 m	A1
7(c)(i)	(magnitude of deceleration =) gradient or Dv / Dt or $15.2 / 4.0$ or equivalent numbers (from graph) in same ratio	C1
	$15.2 / 4.0$ or equivalent numbers in same ratio	C1
	3.8 m/s^2	A1
7(c)(ii)	$(F =) ma$ or $(46 + 9.0) \times 3.8$ or 55×3.8	C1
	210 N	A1
7(c)(iii)	<u>from</u> kinetic energy (and from gravitational potential energy)	B1
	<u>to</u> thermal energy and no intermediate energy	B1
7(d)	due to his mass or mass mentioned	B1
	(his) mass resists a change to its state of motion	B1

4.

Question	Answer	Marks
1(a)	(distance =) area (under graph) or 77×56 or $\frac{1}{2} \times 77 \times 70$ or 4300 or 2700 or $\frac{1}{2} \times 77 \times (56 + 126)$	C1
	$77 \times 56 + (\frac{1}{2} \times 77 \times 70)$ or $4300 + 2700$ or $\frac{1}{2} \times 77 \times (56 + 126)$	C1
	7000 m or $7.0 \times 10^3 \text{ m}$ or 7.0 km	A1
1(b)(i)	$(a =) \Delta v / \Delta t$ or $77 / 70$ (etc.)	C1
	1.1 m/s^2	A1
1(b)(ii)	$(F =) ma$ or $3.8 \times 10^5 \times 1.1$	C1
	$4.2 \times 10^5 \text{ N}$	A1

5.

Question	Answer	Marks
7(a)(ii)	area under curve or formula for area of a triangle / trapezium seen or working shown used	B1
	(accelerating) 80 (m) or (at constant speed) 160 (m) or (decelerating) 32 (m) seen	C1
	(total distance) 270 or 272 m	A1
7(a)(iii)	(average speed) = <u>total</u> distance / (total) time algebraic or numerical	C1
	11 m / s	A1
7(a)(iv)	horizontal line drawn from 0 to 24 s at any level or horizontal line at the value calculated in 7(a)(iii) for at least 16 s	B1
	horizontal line from 0 to 24 s at the value calculated in 7(a)(iii)	B1
7(b)(i)	27	B1
7(b)(ii)	(statement that thinking) distance is proportional to speed or thinking distance increases uniformly with speed or time = (thinking) distance / speed and stated to be equal / not affected or ratio of (thinking) distance to speed is constant or (thinking) distance and speed both increase by same ratio or same change in (thinking) distance for equal changes in speed	C1
	at least two values thinking distance + speed (or inverse) calculated or all values of thinking distance/speed stated to be 0.45 (or inverse 2.2) or proportionality shown by at least two values or double the speed doubles the (thinking) distance or <u>each 20 (km / h) / same</u> increase in speed covers an extra 9 (m) distance	A1
7(b)(iii)	<i>thinking distances</i> increases	B1
	<i>braking distances</i> no change	B1

6.

7(c)	air resistance and weight / gravitational force mentioned	B1
	as speed increases air resistance increases	B1
	<u>resultant</u> / <u>net</u> force decreases or acceleration decreases	B1
	(eventually) air resistance = weight (at terminal velocity) / resultant force is zero	B1
	(at terminal velocity / eventually) zero acceleration / constant speed	B1

7.

Question	Answer	Marks
2(a)	800 N c.a.o.	B1
2(b)(i)	$(F =) ma$ or 80×0.50	C1
	40 N	A1
2(b)(ii)	840 N	B1
2(c)	increasing gradient from origin to $t = 4.0$ s	B1
	decreasing gradient from end of first curve	B1
	line horizontal at origin or at $t = 8.0$ s (and beyond)	B1

8.

Question	Answer	Marks
2(a)	(a vector quantity has) magnitude and direction	B1
2(b)(i)	the wall exerts a force on swimmer	B1
	opposite in direction (to force of swimmer on wall) or to the right	B1
	equal in size (to force of swimmer on wall) or (resultant) force causes acceleration or force on object Y due to X is equal and opposite to force on object X due to Y	B1

Question	Answer	Marks
2(b)(ii)	(water) resistance / resistive force / friction (on swimmer)	B1
	(water) resistance / backward force / resistive force / friction / opposing force increases (with speed / time)	B1
	(at constant speed) forward force is equal to resistive / backward force or no resultant force or forces balance	B1

9.

Question	Answer	Marks
8(a)(i)	weight of horse in mid-section of horse and labelled	B1
	force upwards on at least two legs and labelled	B1
	force downwards on saddle area labelled <i>F</i>	B1
8(a)(ii)	forces equal and opposite	B1
	acts <u>on rider</u> (upwards from horse)	B1

10.

Question	Answer	Marks
1(a)(i)	accelerating or increasing speed	B1
	decreasing acceleration or speed increasing at a decreasing rate or reaches a constant speed	B1
1(a)(ii)	<u>initially</u> acceleration is due to force of gravity / weight (only) or force of gravity / weight larger than air resistance from ($t = 0$ to ≈ 7 s)	B1
	air resistance increases (with speed)	B1
	resultant force decreases / becomes zero or forces balance or air resistance equals weight	B1
1(b)	two different pairs of co-ordinates from <u>straight-line</u> section of graph seen	C1
	correct use of co-ordinates in a division	C1
	48 – 52 m / s	A1

11.

Question	Answer	Marks
2(a)	it / a scalar does not have a <u>direction</u>	B1
2(b)	<u>displacement</u> and <u>force</u> underlined	B1
2(c)(i)	direction (of motion) is changing	B1
	velocity is changing (with time) or it is accelerating (which requires a resultant force)	B1
2(c)(ii)	gravitational attraction / field of the earth	B1

12.

Question	Answer	Marks
7(a)	(between $t = 0$ and $t = 10$ s) it is accelerating or speed / velocity increasing	B1
	between $t = 35$ s and $t = 40$ s, it is decelerating or speed / velocity decreasing	B1
	magnitude of acceleration less than magnitude of deceleration or between $t = 0$ and $t = 10$ s starts from rest / zero speed or between $t = 35$ ms and $t = 40$ s it finishes at rest / with zero speed	B1
7(b)(i)	two pairs of co-ordinates from the straight-line section	C1
	appropriate division using candidate's coordinates	C1
	$11.7 \leq \text{speed} \leq 12.0$ m / s	A1
7(b)(ii)	<u>total</u> distance / <u>total</u> time or (distance =) 390 (m) and 39 (s) $\leq$ (time =) $\leq$ 40 (s)	C1
	9.7 to 10.0 m / s	A1
7(c)(i)	it / velocity has a direction or is a vector or depends on displacement (rather than distance)	B1
7(c)(ii)	answers need not be in this order 1 0 and 10 (s) and forwards / in the direction of movement	B1
	2 21 (s) and 24 (s)	B1
	perpendicular to the motion / to the direction of movement	B1
	3 35 (s) and 40 (s) and backwards / opposite to the direction of movement	B1
7(d)(i)	speed (of bus) varies / changes / decreases / increases	B1
7(d)(ii)	any time on Fig. 1.1 from $10 \text{ s} \leq t \leq 35 \text{ s}$ labelled M	B1

13.

Question	Answer	Marks
2(a)	96 m	B1
2(b)(i)	increases	B1
	time unchanged but speed faster	B1
2(b)(ii)	decreases	B1
	(force of) <u>friction</u> larger	B1

14.

Question	Answer	Marks
1(a)	78 m	B1
1(b)	(starting to) apply the brakes	B1
1(c)(i)	time taken to think / react is the same (and speed is the same)	B1
1(c)(ii)	friction is lower or less backwards / resistant force or car slides or skids / tyres do not grip road / car aquaplanes	B1
	takes longer time to stop or acceleration is less	B1

15.

9(b)(i)	distance travelled before brakes applied (after seeing emergency)	B1
9(b)(ii)	any 2 of - poor tyre surface - poor road conditions, e.g. rain, ice - high speed (of car) or good aerodynamic car shape or tail wind - large mass / weight of car - driver applies less force to pedal / less braking force	B1B1

16.

Question	Answer	Marks
9(a)(i)	amount / measure / quantity of matter / substance (in a body) or (property of a body) that resists (change in) motion	B1
9(a)(ii)	force / pull of <u>gravity</u> or in a gravitational field	B1
9(b)	(a =) F / m algebraic or numerical or 400 (N) seen	C1
	400 / 60	C1
	6.7 m/s ²	A1
9(c)(i)	initial curve of correct shape from origin at 0 to 30 s	B1
	horizontal line at 44 m/s from 30 s to 40 s	B1
	speed drops from 44 to 5 m/s in a time of 4 s	B1
	constant speed of 5 m/s until 200 s (and then to zero)	B1
9(c)(ii)	(a =) $(v - u) / t$ algebraic or numerical	C1
	9.7 or 9.8 m/s ²	A1
9(d)(i)	upwards force / air resistance = weight / downward force	B1
9(d)(ii)	increase in air resistance / upwards force (from parachute)	B1
	upwards / force / air resistance larger than weight / downwards force or resultant force upwards	B1
9(d)(iii)	air resistance decreases (as speed reduces) until it equals weight or (with parachute open) air resistance = weight at lower speed	B1

17.

(a)	$(F =) ma$ or 35×2.6	C1
	91 N	A1
(b) (i)	straight line from origin to dashed line / t_1	B1
	gradually decreasing gradient until the line is horizontal	B1
(b) (ii)	area <u>under</u> the line or area between line and x-axis	B1

18.		
(a)	(i) equal and opposite forces or no resultant force	B1
	(ii) moves / slides	B1
	constant velocity or constant speed and direction	B1
(b)	(resultant force) 15 (N) seen	C1
(b)	(a=) F / m in any form numerical or algebraic	C1
	$0.60 \text{ m} / \text{s}^2$	A1
19.		
(a)	A – contact or reaction force B – driving force C – force of gravity D – air resistance and friction	B1
(b)	(i) 800 kg	B1
	(ii) 400 N	B
	(iii) (a =) F / m formula or with numbers	C1
	$0.50 \text{ m} / \text{s}^2$	A1
(c)	change in $v = at$ in words or numbers	C1
	$12 \text{ m} / \text{s}$	A1
20.		
(a)	kinetic energy at start	B1
	thermal energy/heat energy/ internal energy at end	B1
(b)	(i) $0.4(0) \text{ s}$	B1
	(ii) (d=) $s \times t$ numerical or algebraic	C1
	2.8 m	A1
	(iii) area under graph (between 0.4 and 2.4 s) or time (difference) $\times$ <u>average</u> speed or $\frac{1}{2} \times$ time (difference) $\times$ initial speed	B1
(iv)	horizontal line from (0,5) to (0.4,5)	B1
	line showing braking with same gradient as original line	B1

- (v) less friction B1
 less deceleration **or** graph less steep B1
or less force backwards/less force opposing motion
or same KE lost/work done by friction
 longer time to stop B1
or larger area under (speed-time) graph
or work = force x distance applied correctly

21.

- (a) (i) **D and either** lorry accelerates (forward) **or** resultant force is forward [B1]
 (ii) air resistance **or** (air) drag **or** friction (between tyres and road) [B1]
 (b) (i) 30 000 kg [B1]
 (ii) (a=) F/m algebraic in any form or numerical [C1]
 $0.5(0) \text{ m/s}^2$ [A1]
 (c) direction or velocity is changing **or** acceleration or force is sideways or towards centre (of circle) [B1]

22.

- (a) (i) 850 N B1
 (ii) $KE = PE/mgh$ **or** $mgh = 5.5 \times 10^4$ C1
 $65/64.7(0588) \text{ m}$ A1
 (b) $WD = Fx$ **or** KE/x **or** $5.5 \times 10^4/33$ **or** $v = 35(.97)$ **and** $a = 19(.60)$ **and** $F = ma$ C1
 $1700/1670/1667/1666.7 \text{ N}$ A1

23.

- (a) (i) speed and mass B2
 (ii) 1 speed and direction B1
or distance/time and direction
or displacement/time
 2 direction changes B1
 (iii) force of gravity from/towards Earth B1
 force is centripetal B1
or at right angles to motion/velocity
 (b) (i) 450 000 N B1
 (ii) (a =) F/m or 50 000/40 000 C1
 1.25 m/s^2 A1

- 24.
- (a) a force B1
when objects slide over/rub one another
OR opposes (relative) **motion/movement** B1
- (b) (i) constant/uniform speed OR constant/uniform velocity OR zero acceleration B1
- (ii) ($F =$) ma seen in any form numerical or algebraic C1
1200 (N) OR 6200 N seen C1
3800 N A1
- (iii) Force B increases OR backwards force/resistance/friction/drag increases M1
as speed/velocity increases A1
- 25.
- (a) **at start:** chemical (potential) energy B1
at end: PE/GPE/gravitational energy B1
at end: heat/thermal/internal energy B1
at end: KE **or** intermediate KE from chemical energy B1
- (b) (i) 0 B1
- (ii) it increases B1
to constant value B1
- (iii) gradient **or** $(v-u)/t$ **or** $(1400-600)/40$ **or** other correct numbers C1
 20 m/s^2 A1
- (iv) ($F =$) ma **or** $1.6 \times 10^6 \times 20$ C1
 $3.2 \times 10^7 \text{ N}$ A1
- (v) $4.8 \times 10^7 \text{ N}$ B1
- (c) (i) to every action there is an equal and opposite reaction
or forces act in pairs of equal size **and** in opposite directions/on
different bodies B1
- (ii) downward force on gas B1
equal and opposite to upward/(b)(v) force (on rocket) B1
- 26.
- (a) (i) X weight // (force of) gravity // gravitational (force)
and Y air resistance // (air) drag // wind resistance // **air** friction B1
- (ii) (Y) opposes motion // diver moves down // air molecules hit faster from below B1
- (b) (i) accelerates // falls faster // speed/velocity greater B1
X is larger than Y // resultant/net force downwards B1
- (ii) **two** of:
X stays the same // Y increases // forces become equal (and opposite) B2

- 27.
- (a) gravitational/centripetal (pull/attraction) of the **Sun** B1
B1
- (b) (i) arrow touching Venus towards centre/left (must pass through Sun if extended) B1
- (ii) $(F =) ma$ or $4.9 \times 10^{24} \times 9.7 \times 10^{-3}$ C1
 $4.8(4.753) \times 10^{22} \text{ N}$ A1
- (c) direction of movement $\perp$ direction of force C1
force/it is perpendicular/at right angles to distance moved **or** does not move **any** distance in direction of force A1
- 28.
- (a) (i) distance travelled while thinking/in reaction time or before braking starts B1
- (ii) distance travelled while brakes applied/car decelerates B1
- (b) (i) speed (of cars) or same force/pressure on pedal or same braking force or same tyres or condition of brakes B1
- (ii) greater inertia/kinetic energy/momentum or smaller deceleration/acceleration B1
- (c) any road condition, e.g. icy, wet, poor surface, slippery/smooth/rough surface and its correct effect on distance B1
correct explanation that refers to friction e.g. more friction when dry B1
- (d) pressure low(er) (with larger area) B1
- (e) (i) $a = v/t$ any algebraic or numerical value e.g. 20/4; 20/3.4; 20/4.6; 20/0.6 C1
 $5(.0)\text{m/s}^2$ A1
- (ii) $F = ma$ algebraic or numerical e.g. $900 \times$ (i) C1
4500 N e.c.f. (i) A1
- (iii) correct axes labelled with quantity and/or unit B1
horizontal line at 20 m/s from 0 to 0.6 s B1
straight line from end of horizontal section or from (0.6,20) to (4.6,0) or (4,0) B1
- (iv) area (under graph or of trapezium) B1